

Instructions:

You've been assigned to remotely investigate a compromised Linux server as part of a live incident response scenario. The affected system has been imaged and all necessary forensic artifacts have been gathered. Your objective now is to investigate the scope of the compromise and identify any persistence mechanisms installed by the attacker.

Use your endpoint analysis skills to conduct the investigation and answer the questions below. Remember, many investigative commands will need to be run with sudo to return comprehensive results.

Challenge File:

03_Endpoint_Security/Linux/Challenges/challenge.elf

To start the challenge, run `sudo ./challenge.elf` in a terminal window. Make sure to not close the program or window until after you've completed the challenge. However, you can temporarily minimize it. After running the executable, you can open additional terminal sessions to complete the investigation.

After completing the challenge, you can clean up and restore the system by terminating the challenge.elf process (Press Ctrl + C).

Q1 Simulate the compromise by running `sudo ./challenge.elf` in a terminal window. Make sure to not close the program or window until after you've completed the challenge. However, you can temporarily minimize it. Enter "done" to continue.

Done

Q2 What port is the malware listening on?

31337

Q3 What is the Process ID (PID) of the malware? This answer cannot be verified, but it's required for the next question(s).

40095

Q4 What is the name of the binary or executable associated with the network listener?

Kitty.meow

Q5 What is the full path of the previously identified binary?

/tmp/kitty.meow

Q6 Run the ps command against the previously identified malicious Process ID (PID). What is the full command line invocation of this binary?

/tmp/kitty.meow -l -p 31337

Q7 List the full process tree for the malicious process. What is the name of the parent process that spawned the malware?

systemd

Q8 What is the SHA-256 hash value of the malware? Note: this is not asking about challenge.elf.

de49cac83b4645476865e1faa39ac6c9e833baa1ea26fde02ec8045ec71a160d

Q9 Extract all printable string characters from the binary. Review the results for any indicators such as risky objects, descriptions, comments,

banners, or usage information. Based on these findings, what utility or program can you identify?

nc

Q10 Examine the system-wide cron jobs configured on the local system. Identify any entries that may be malicious, indicating persistence mechanisms or continued compromise by the attacker. What is the full file path of the script that is executed?

/tmp/exfiltr8.sh

Q11 Use a service like crontab.guru to interpret the cron schedule. What is the specific day when this cron entry runs?

October 13

Q12 Now, list your own user account's crontab entries. What IP address does the attacker's beacon reach out to?

no crontab for root

Q13 Excellent work! Clean up and restore the system by terminating the challenge.elf process (Press Ctrl + C). Afterwards, enter "done" to continue.

done

Challenge URL

<https://challenges.malwarecube.com/#/c/65889869-d947-44d8-9241-a859a07ace91>